FACULTY OF ENGINEERING OF THE UNIVERSITY OF PORTO

Systems Engineering

Team Meeting Report

SWAID — Plate Resonance (PR)

Meeting Date:	March 18, 2026
Location:	FEUP I-108
Time:	16:10 — 17:45
Proposed by:	Professor Sérgio Reis Cunha

1 Attendance & Logistics

Team Member	Present	Role / Focus
Miguel Amorim	[P]	Team Leader
João Oliveira	[P]	Technical Leader (HW)
Artur Kovbasa	[P]	Technical Leader (SW)
Afonso Cruz	[P]	Member
Alexandre Gorra	[P]	Member
Beatriz Martins	[F]	Member
Hugo Gomes	[P]	Member
João Pinto	[P]	Member
Lucas Aparicio	[P]	Member

2 Meeting Objectives & Discussion Notes

2.1 Presentation Planning

Discussion & Notes:

- Define the workflow for the upcoming 12-minute classroom presentation, prioritizing script development and slide design.
- Assign roles for the presentation: Miguel, Beatriz, Lucas, and João are responsible for the script and speaking; Thomaz and Cruz are assigned to slide creation.
- Establish a strict timing breakdown: 8 minutes for the core presentation, 2 minutes for Q&A, and a 2-minute contingency margin.

Conclusions:

- We will make a quizz based presentation and further discuss interactive ways to teach the topics.
- We will start by doing research on the topic and define the main topics to present, so then we can write a script for the presentation
- Once the script is done, Afonso e João Thomaz will elaborate some slides to go along with the presentation

2.2 Simulation Development and Parameter Optimization

Discussion & Notes:

- Prepare comparative simulation results for the Wednesday review, focusing on two primary options: Square Aluminum and Rectangular Steel.
- Prioritize the integration of the amplitude variable within the simulation environment to increase result fidelity.

- Execute automated sweeps for various transducer configurations;
- Investigate the following technical parameters:
  - The specific impact of the damping factor on pattern formation.
  - The total number of achievable precision modes.
  - Implementation of forced and individual visualization modes for better symbol analysis.

Conclusions:

- Amplitude variable has been integrated into the simulator. We must now test how it affects the symbol generation. As noted by the frequency response graph, higher frequency modes are highly attenuated (<-60dB). We must understand how much power should be needed to drive each symbol and translate that to an amplitude value (volume control).
- It is imperative to reach a plate choice until next week, therefore we must make a lot more simulations with different setups with our current plate options (should we opt for a larger and thinner plate or a smaller and thicker?)
- Currently the automatic sweep is generating a lot of repeated symbols, we should make a filter to save only the best unique symbols for each configuration.
- We must experiment a lot more transducer placement variations.
- Is it worth to explore exotic parameter configurations (different frequencies between transducers, different amplitudes, phase differences that don't sum to 2π) ?

2.3 Sound Card and Software API Architecture

Discussion & Notes:

- Define the functional requirements for the Software API, ensuring the sound card driver remains independent of the specific Chladni symbol alphabet.
- Design the driver to receive hardware parameters (frequency, phase, and amplitude) for each channel via USB.
- Develop the API as a modular library (C/C++ or Python) for the HI (Human Interface) team to use.
- Propose a high-level command structure, such as:
  - `gen_chladni(uint symbol)`: Maps a symbol ID to specific 4-channel parameters.
  - `free_chladni(var)`: Provides granular control over individual channel variables.
- Ensure the API architecture integrates control logic for lighting effects, specifically managing the LED Matrix and side-mounted LED strips.

Conclusions:

- We have made an update to the sound card software and will be testing in the lab if we can control well channel independence.
- Overall system requirements will be further discussed in a inter team meeting to align the project development across all teams.

2.4 Weekly Deliverable: Analysis and Functional Architecture

Discussion & Notes:

- Outline the framework for Deliverable #4, focusing on the ”Analysis and Functional Architecture” of the Plate Resonance system.

Conclusions:

- Artur, Alexandre and Hugo are responsible with this document.

2.5 Inter-Team Meeting

- A inter team meeting must be scheduled as soon as possible and professors must be notified of its occurence and sent afterward a meeting report outlining the topics discussed.
- The objectives of this meeting are:
  - Define a concrete and detailed system architecture and a global requirements list for the whole project
  - Update and finish the Bill of Materials
  - Establish a development cycle for the project

3 Task Assignment

*Categories: Management (**G**), Deliverables (**E**), Development (**D**).*

Task Description	Cat.	Assigned To	Status	Deadline
Deliverable #4	E	Artur, Alexandre, Hugo	Open	20/3
Team Presentation Script	E	Beatriz, João, Lucas, Miguel	Open	22/3
Team Presentation Slides	E	Afonso, João Thomaz	Open	22/3
Chladni Simulations	D	Artur , Alexandre, Hugo	Open	23/3
SoundCard testing	D	Afonso, João Thomaz	Open	23/3
SWAID meeting - Global requirements document	G/D	Miguel + PR Team Review	Open	23/3
Software Abstraction Layer architecture	D	PR TEAM	Open	29/3

4 Next Meeting

- **Proposed Date:** March 25, 2026
- **Proposed Location:** FEUP I-108

4.1 Next week's main objectives:

- Have a good symbol alphabet candidates
- Select a plate (material and dimensions) to order
- Close the Bill of Materials
- Have a good presentation